

Sports Analytics Dashboard and Match Outcome Prediction Using Machine Learning

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Undergraduate Final Project Report

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Sports Analytics and Machine Learning

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Approval Page

This mini thesis entitled Sports Analytics Dashboard and Match Outcome Prediction Using Machine Learning has been prepared as an undergraduate final project report. The work describes the implementation, evaluation, and results of the sports analytics platform developed in the repository.

This report is submitted for academic review and final project assessment.

Approved by:

Academic Supervisor: _____

Date: _____

Declaration

I declare that this mini thesis is based on the implementation found in the Sports Analytics Dashboard project repository. The dataset description, model results, screenshots, and discussion are taken from the actual application files, generated outputs, and saved model bundle. No experimental result or screenshot has been invented.

All external references are listed in APA style in the References section.

Acknowledgements

I would like to thank the academic supervisor and lecturers who provided guidance during this final project. I also appreciate the open-source communities behind Python, pandas, scikit-learn, Streamlit, Plotly, and OpenFootball, because these tools and datasets made the project possible.

This project helped me understand how software engineering, data analysis, machine learning, and user interface design can be combined into one complete application.

Abstract

Football is one of the most popular sports in the world, and match data can be used to understand team performance and support prediction. This project developed a Sports Analytics Dashboard and Match Outcome Prediction system using Python, Streamlit, Plotly, pandas, NumPy, scikit-learn, Joblib, and Pillow. The application uses verified football data from the OpenFootball public dataset cache and stores a processed dataset containing 684 match records, 27 columns, and 23 real football clubs.

The system includes dataset overview, data visualization, team comparison, match prediction, World Cup prediction, model performance, and project information pages. The saved model bundle uses Logistic Regression for match outcome classification. The model reached an accuracy of 49.01%, precision of 48.16%, recall of 49.01%, and F1 score of 48.53% on the test split. The prediction page also supports verified teams, optional live API providers, cached data, probability output, confidence score, and explanation text.

The result is a working undergraduate capstone application that demonstrates the full process of collecting football data, preparing features, training a model, evaluating performance, and presenting the results through a modern web dashboard.

Keywords: sports analytics, football analytics, machine learning, Streamlit, match prediction, data visualization

Table of Contents

Approval Page

Declaration

Acknowledgements

Abstract

List of Figures

List of Tables

Chapter 1 Introduction

Chapter 2 Literature Review

Chapter 3 System Design and Methodology

Chapter 4 Implementation and Results

Chapter 5 Conclusion

References

Appendices

List of Figures

Figure 4.1. Home page of the Sports Analytics Dashboard.

Figure 4.2. Dataset Overview page showing dataset statistics and table preview.

Figure 4.3. Correlation heatmap generated from the football dataset.

Figure 4.4. Data Visualization page with interactive charts.

Figure 4.5. Team Comparison page with head-to-head and performance charts.

Figure 4.6. Match Prediction page with league and team selection.

Figure 4.7. Prediction output with probability and explanation.

Figure 4.8. World Cup Prediction page with team strength index.

Figure 4.9. Model Performance page showing evaluation metrics.

Figure 4.10. Confusion matrix section from the Model Performance page.

Figure 4.11. Feature importance section from the Model Performance page.

Figure 4.12. About Project page explaining the technology stack and pipeline.

List of Tables

Table 3.1. Dataset summary used in the project.

Table 3.2. Main engineered features used by the prediction pipeline.

Table 4.1. Model evaluation metrics.

Table 4.2. Top feature importance values.

Table 4.3. Confusion matrix from the saved model bundle.

Table C.1. Sample rows from the football dataset.

CHAPTER 1 INTRODUCTION

1.1 Background

Football produces a large amount of data from every match, such as goals, results, team form, and performance trends. In the past, this information was often read manually from tables or match reports. Today, sports analytics allows the same information to be processed and displayed in a more useful way. A dashboard can help users see patterns faster, and machine learning can be used to estimate possible match outcomes.

This project focuses on building a complete football analytics platform. It is not only a model script, and it is not only a dashboard. The system combines dataset processing, data visualization, team comparison, prediction, model evaluation, and a World Cup strength index into one Streamlit web application.

1.2 Problem Statement

Football data is available from different sources, but it is not always easy for students or general users to explore it. Raw datasets can be difficult to read, and prediction outputs can be hard to understand if they only show a class label without probability or explanation. Another problem is that many simple dashboard projects use artificial data, which makes the application less realistic.

The problem addressed in this project is how to build a realistic, usable, and explainable sports analytics dashboard using verified football data and a machine learning pipeline.

1.3 Research Questions

- How can verified football match data be organized into an interactive analytics dashboard?
- Which football features can be used to support match outcome prediction?
- How well does the saved machine learning model perform on the available dataset?
- How can prediction results be explained in simple English for dashboard users?

1.4 Objectives

- To develop a Streamlit dashboard for football dataset exploration and visualization.

- To create a team comparison feature using real football clubs and performance indicators.
- To build a match prediction workflow that returns class probabilities and confidence score.
- To evaluate the model using accuracy, precision, recall, F1 score, confusion matrix, and feature importance.
- To include a World Cup prediction page based on a transparent weighted strength index.

1.5 Scope

The project uses the existing repository as the source of truth. The main dataset is the processed football match dataset saved in `dataset/football_matches.csv`. The dashboard supports the implemented pages and the optional live provider structure already present in the code. The model result discussion is limited to the saved model bundle and does not claim performance beyond the available dataset.

1.6 Benefits

The project benefits students by showing how a machine learning model can be connected to a real web application. It also helps users understand football data through charts, tables, and simple prediction explanations. From a software engineering point of view, it demonstrates caching, modular code, reusable components, and separation between data utilities, model logic, visualization, and pages.

1.7 Report Organization

Chapter 1 introduces the background, problem, questions, objectives, scope, and benefits. Chapter 2 reviews sports analytics, football analytics, machine learning, dashboards, and related tools. Chapter 3 explains the system design and methodology. Chapter 4 presents the implementation and results from the actual application. Chapter 5 concludes the report and explains limitations and future work.

CHAPTER 2 LITERATURE REVIEW

2.1 Sports Analytics

Sports analytics is the use of data to understand performance, strategy, and decision-making in sports. In football, analytics can support scouting, team preparation, post-match review, and prediction. The value of analytics is not only in collecting data but also in turning data into useful insight.

2.2 Football Analytics

Football analytics focuses on match events, team performance, scoring patterns, defensive strength, and player or team behavior. Public football datasets make it possible for students to build realistic projects without depending only on private professional data.

2.3 Machine Learning

Machine learning allows a computer program to learn patterns from historical data. In this project, the task is classification because the model predicts one of three possible outcomes: Home Win, Draw, or Away Win.

2.4 Data Visualization

Data visualization helps users understand complex data quickly. Charts such as histograms, boxplots, heatmaps, radar charts, and bar charts are useful because they show patterns that are difficult to see in raw tables.

2.5 Random Forest

Random Forest is an ensemble method introduced by Breiman (2001). It combines many decision trees and is widely used for classification tasks. In this project, Random Forest is included in the model comparison pipeline, while the saved best model in the current bundle is Logistic Regression.

2.6 Streamlit Dashboard

Streamlit is a Python framework for building data applications. It is useful for undergraduate projects because it allows the developer to connect Python analysis, charts, machine learning models, and user interface components in one application.

2.7 Match Prediction Systems

A match prediction system estimates the possible result of a future or selected match. A good system should not only show the predicted outcome but should also show probability, confidence, and explanation so the user understands the reason behind the result.

2.8 Related Studies

Previous work in sports prediction shows that machine learning can support result prediction when meaningful features are available. However, model performance depends strongly on dataset quality, feature design, and the nature of the sport. Football is difficult to predict because draws, injuries, tactics, and random events can influence the final result.

The literature supports the design direction of this project: use verified data, prepare meaningful features, evaluate the model honestly, and present the results through a dashboard that is easy to understand.

CHAPTER 3 SYSTEM DESIGN AND METHODOLOGY

3.1 Project Overview

The project is a Python-based web application built with Streamlit. The application is organized into `app.py`, `pages`, `components`, `assets`, `models`, `dataset`, and `utils` folders. The `pages` folder contains dashboard pages, while `utils` contains data loading, live provider handling, model utilities, report generation, visualization utilities, training logic, and World Cup indexing.

3.2 Dataset

Table 3.1. Dataset summary used in the project.

Item	Value
Rows	684
Columns	27
Unique teams	23
Leagues	Premier League
Seasons	2024, 2025
Missing values	0
Duplicate rows	0
Data source	openfootball public EPL Football.TXT cache

The dataset summary in Table 3.1 comes directly from `dataset/football_matches.csv`. The project currently contains real Premier League data from the OpenFootball public cache.

3.3 Data Collection

The data collection logic is implemented in the live data utilities. The priority is to use optional live APIs when API keys are available, then cached API data, then OpenFootball public data, then `football-data.co.uk` public data, and finally verified real team lists. The application does not fabricate teams or fixtures.

3.4 Data Cleaning

The project parses football match data, standardizes team names, removes duplicate match rows, converts dates, and ensures that required columns such as home team, away team, goals, result, season, and league are available. Missing values and duplicate rows are shown on the Dataset Overview page.

3.5 Feature Engineering

Table 3.2. Main engineered features used by the prediction pipeline.

Feature	Description
home_team_strength	Estimated home team strength before the match.
away_team_strength	Estimated away team strength before the match.
home_recent_form	Recent form signal for the home team.
away_recent_form	Recent form signal for the away team.
home_avg_goals	Average goals signal for the home team.
away_avg_goals	Average goals signal for the away team.
home_defense_rating	Defensive rating for the home team.
away_defense_rating	Defensive rating for the away team.
team_strength_delta	Difference between home and away team strength.
goal_power_delta	Difference between attacking goal signals.

3.6 Machine Learning Pipeline

The machine learning pipeline is implemented in `utils/train_model.py`. It loads verified football match data, engineers model features, encodes the target variable, splits the dataset into training and testing sets, trains candidate models, compares metrics, and saves the selected bundle using Joblib.

3.7 Model Training

The saved model bundle reports Logistic Regression as the selected model. The target variable is result, with the classes Away Win, Draw, and Home Win. The training split contains 533 rows and the testing split contains 151 rows.

3.8 Model Evaluation

The model is evaluated using accuracy, precision, recall, F1 score, cross-validation mean and standard deviation, classification report, confusion matrix, ROC curve, and feature importance. These values are displayed on the Model Performance page.

3.9 Dashboard Development

The dashboard uses Streamlit for the web interface and Plotly for interactive charts. Shared styling and navigation are handled in components/ui.py. This keeps the visual identity consistent across pages.

3.10 Prediction Workflow

For club matches, the user selects a league and either chooses an available upcoming fixture or manually selects two verified real teams. The system calculates team statistics, recent form, team strength, and probabilities. For World Cup prediction, national teams are compared using a weighted strength index.

3.11 System Architecture

The system workflow is: verified data source, data cleaning, feature engineering, model training, prediction utility, Streamlit dashboard, and downloadable reports. This architecture separates data processing, model logic, visual components, and page-level user interface code.

CHAPTER 4 IMPLEMENTATION AND RESULTS

4.1 Implementation Overview

The implementation is divided into reusable Python modules. The `app.py` file provides the home page. The `pages` folder contains the user-facing dashboard pages. The `utils` folder contains data loading, live provider logic, model prediction, report generation, visualization functions, model training, World Cup calculations, and presentation content helpers.

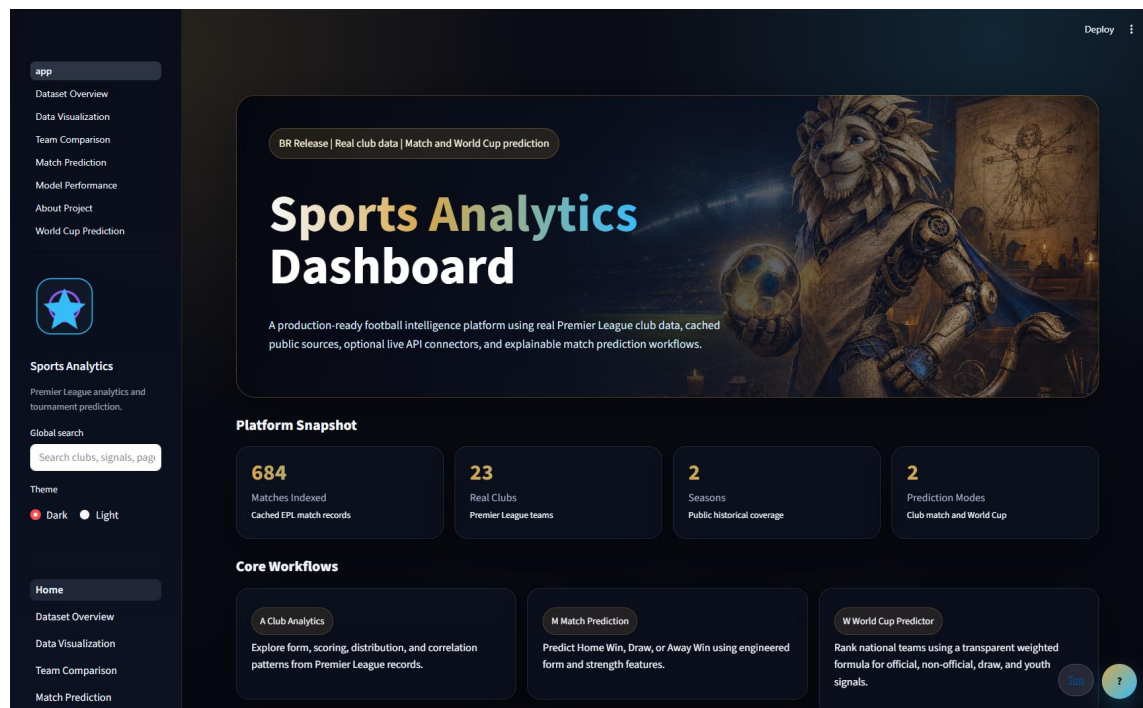


Figure 4.1. Home page of the Sports Analytics Dashboard.

4.2 Dataset Overview Page

Figure 4.2 shows the Dataset Overview page developed in this project. It displays dataset size, missing values, duplicate values, an interactive data table, dataset summary, feature types, and a download button.

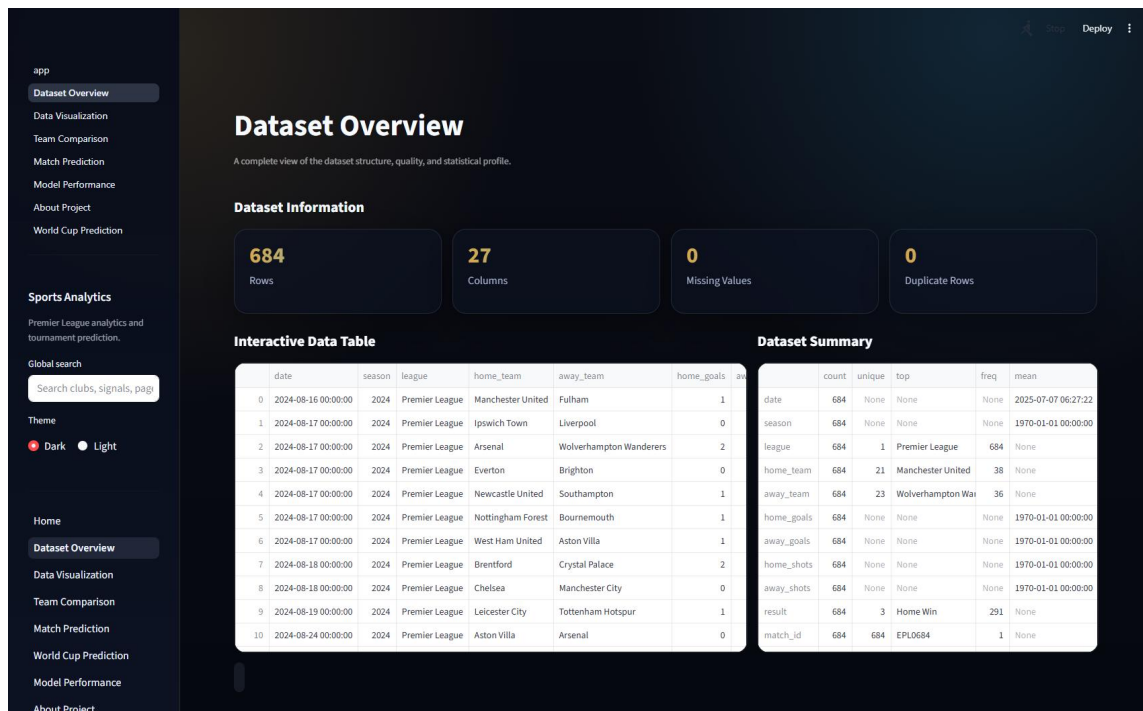


Figure 4.2. Dataset Overview page showing dataset statistics and table preview.

Figure 4.3 shows the correlation heatmap. This chart helps identify how numerical features relate to each other before model training and evaluation.

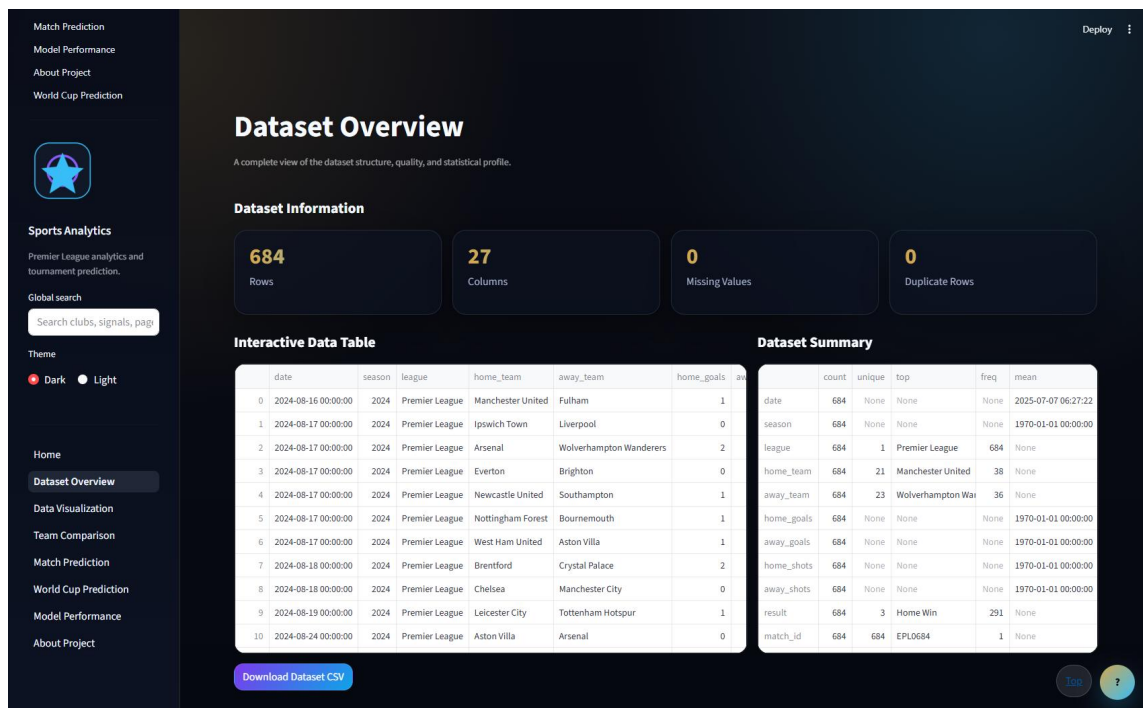


Figure 4.3. Correlation heatmap generated from the football dataset.

4.3 Data Visualization Page

Figure 4.4 shows the Data Visualization page. The page includes interactive charts such as outcome distribution, correlation matrix, histograms, boxplots, scatter plots, and violin charts. Users can filter by season, league, and team.

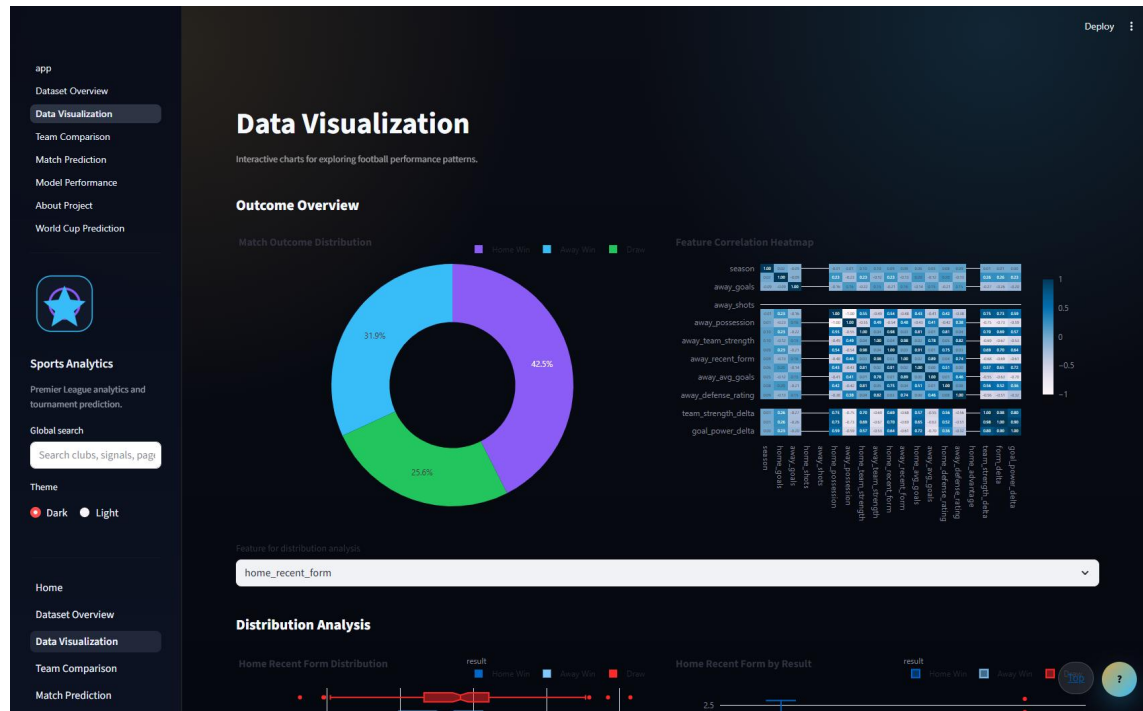


Figure 4.4. Data Visualization page with interactive charts.

4.4 Team Comparison

Figure 4.5 shows the Team Comparison page. It allows the user to choose two real football clubs and compare win rate, goals, average performance, radar chart, bar charts, head-to-head data, and performance trend.

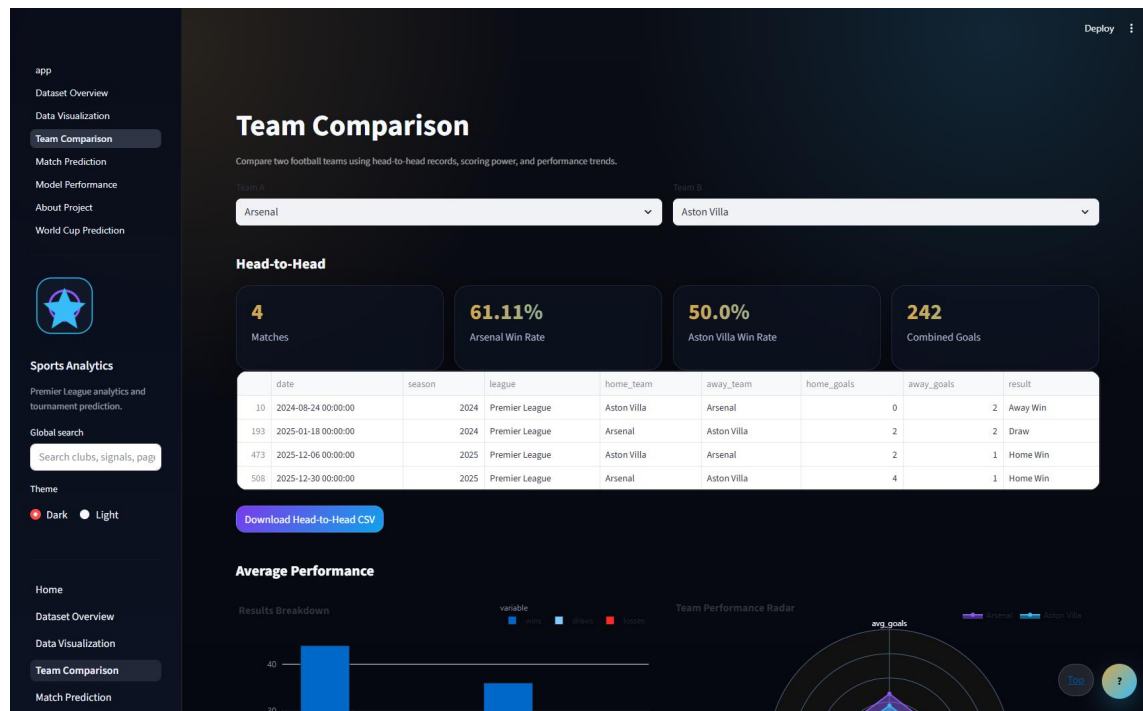


Figure 4.5. Team Comparison page with head-to-head and performance charts.

4.5 Match Prediction

Figure 4.6 shows the Match Prediction page. The page includes league selection, upcoming match selection when verified fixtures are available, and manual team selection using only real teams.

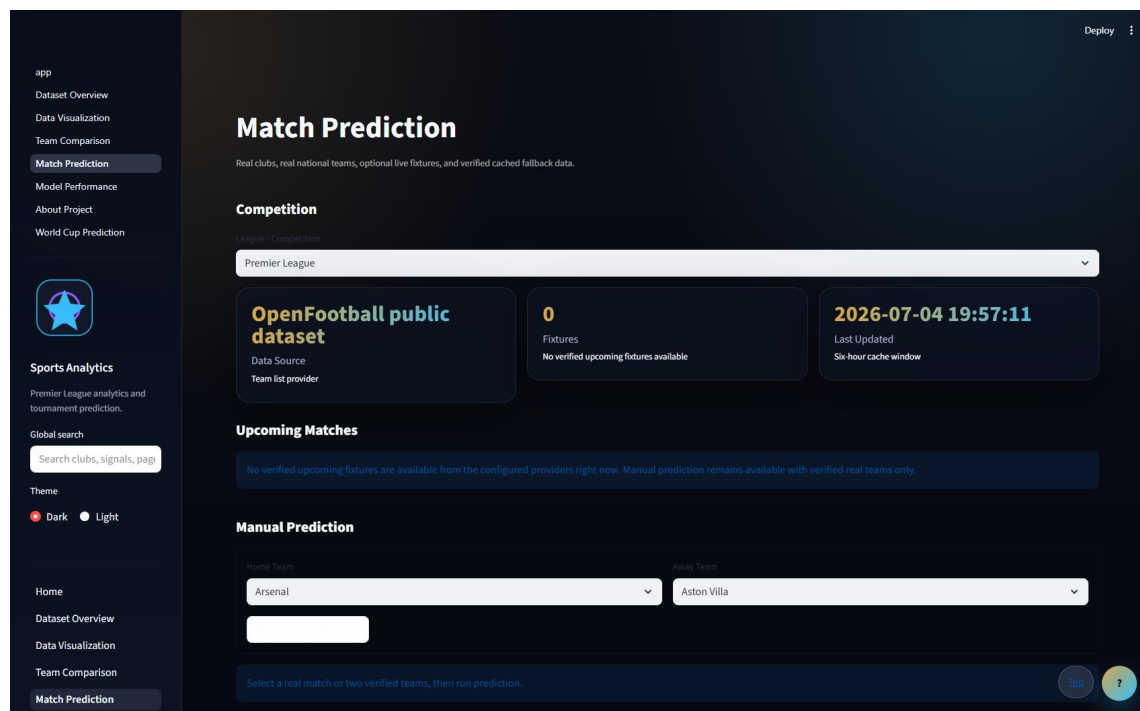


Figure 4.6. Match Prediction page with league and team selection.

Figure 4.7 shows a prediction output generated from the running application. The output contains the predicted outcome, confidence score, provider badge, recent form cards, strength comparison, probability bars, and explanation.

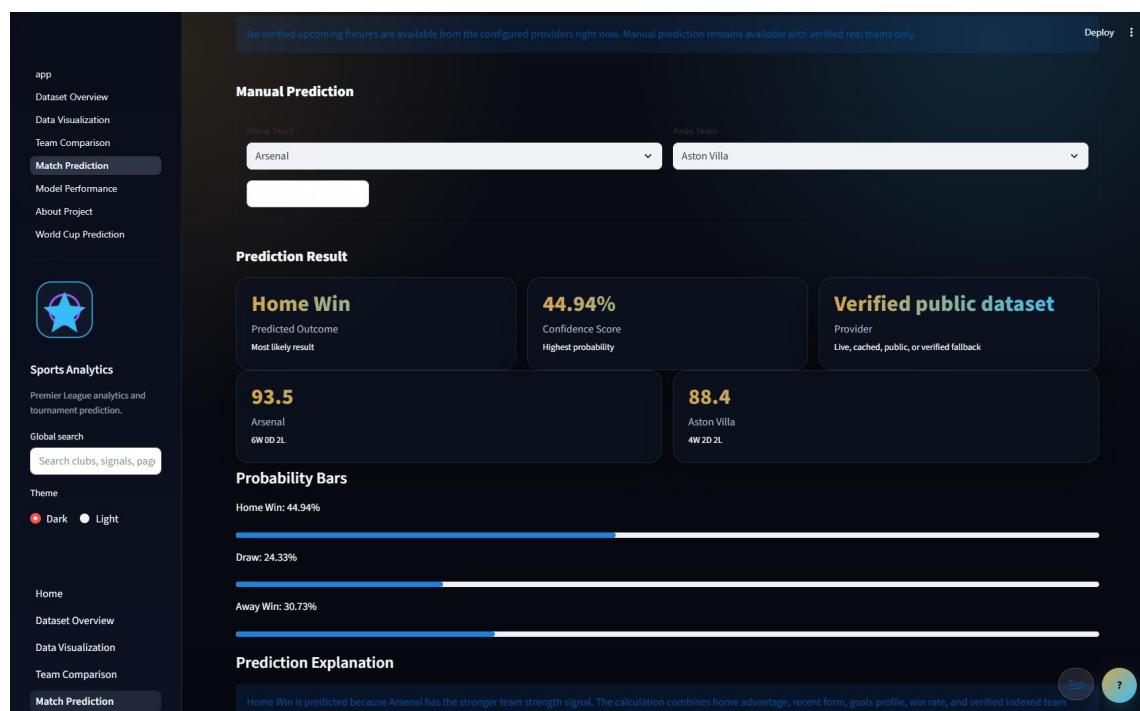


Figure 4.7. Prediction output with probability and explanation.

4.6 World Cup Prediction

Figure 4.8 shows the World Cup Prediction page. It uses a transparent weighted formula: official senior win is 1.00, draw is 0.50, friendly or non-official win is 0.30, and youth signal is 0.20. The result is shown as a team strength index.

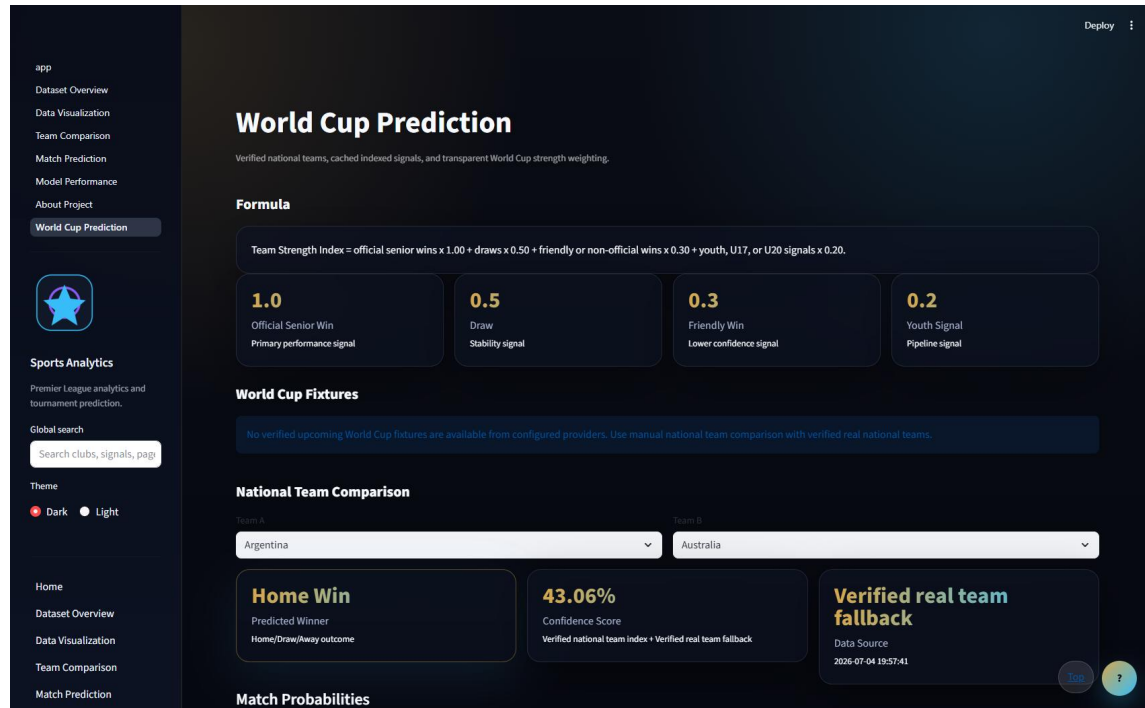


Figure 4.8. World Cup Prediction page with team strength index.

4.7 Model Performance

Table 4.1. Model evaluation metrics.

Metric	Value (%)
Accuracy	49.01
Precision	48.16
Recall	49.01
F1	48.53
Cv Mean	45.45

Cv Std	1.74
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The saved model is Logistic Regression. Table 4.1 reports the actual metrics from the saved model bundle. The model reached 49.01% accuracy and 48.53% F1 score. These values show that the model works as a baseline classifier but still has room for improvement.

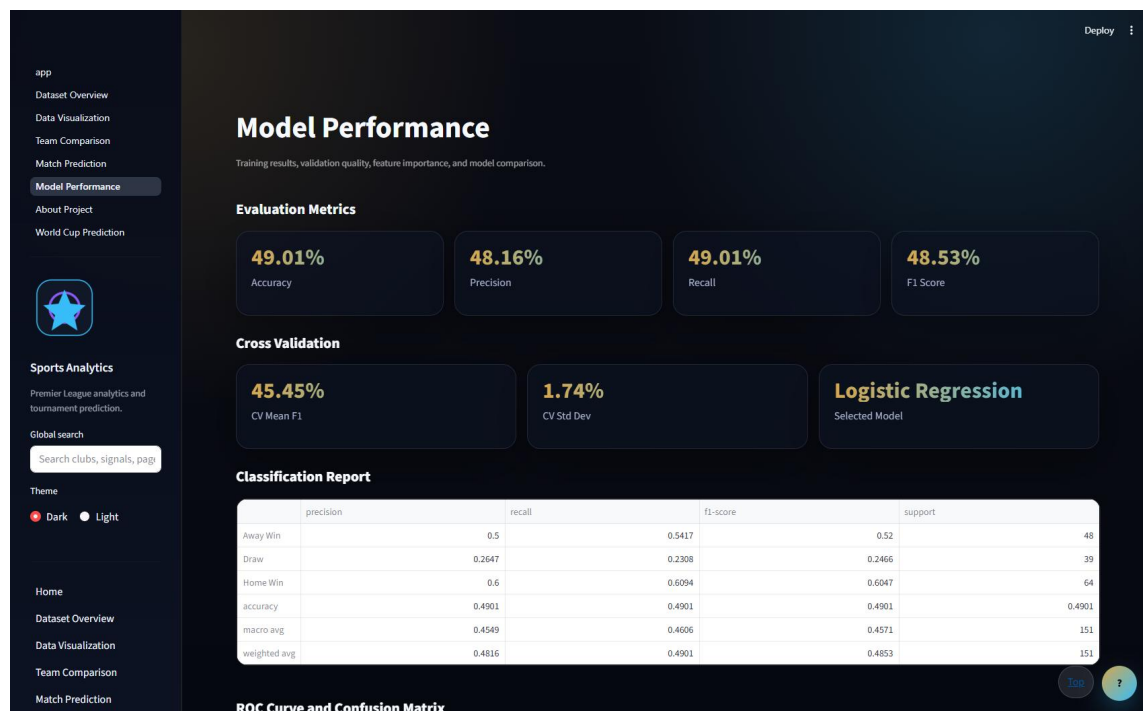


Figure 4.9. Model Performance page showing evaluation metrics.

Figure 4.10 shows the confusion matrix section from the Model Performance page. The confusion matrix helps show which outcome classes are easier or harder for the model to classify.

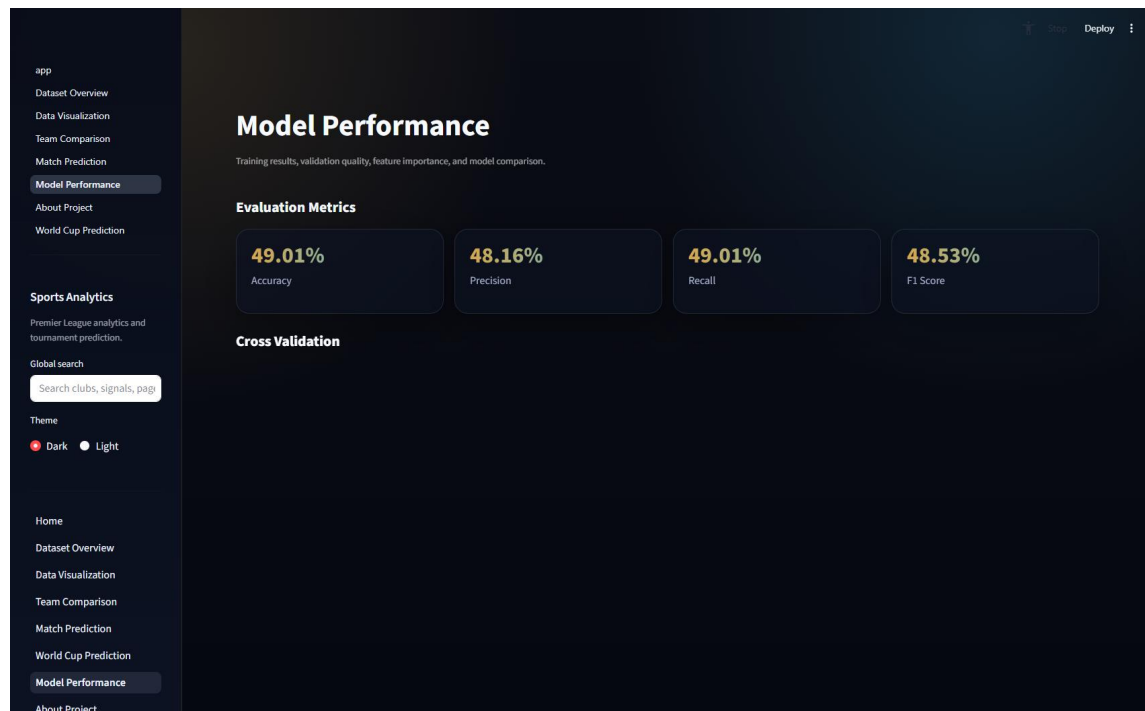


Figure 4.10. Confusion matrix section from the Model Performance page.

Table 4.2. Top feature importance values.

Feature	Importance
away_defense_rating	0.496
home_team_strength	0.3564
home_defense_rating	0.304
home_avg_goals	0.297
away_avg_goals	0.2726
away_team_strength	0.2418
home_recent_form	0.1297
team_strength_delta	0.1246

Table 4.2 and Figure 4.11 show the most important features from the current saved model bundle. The highest ranked feature is away_defense_rating, followed by home_team_strength and home_defense_rating.

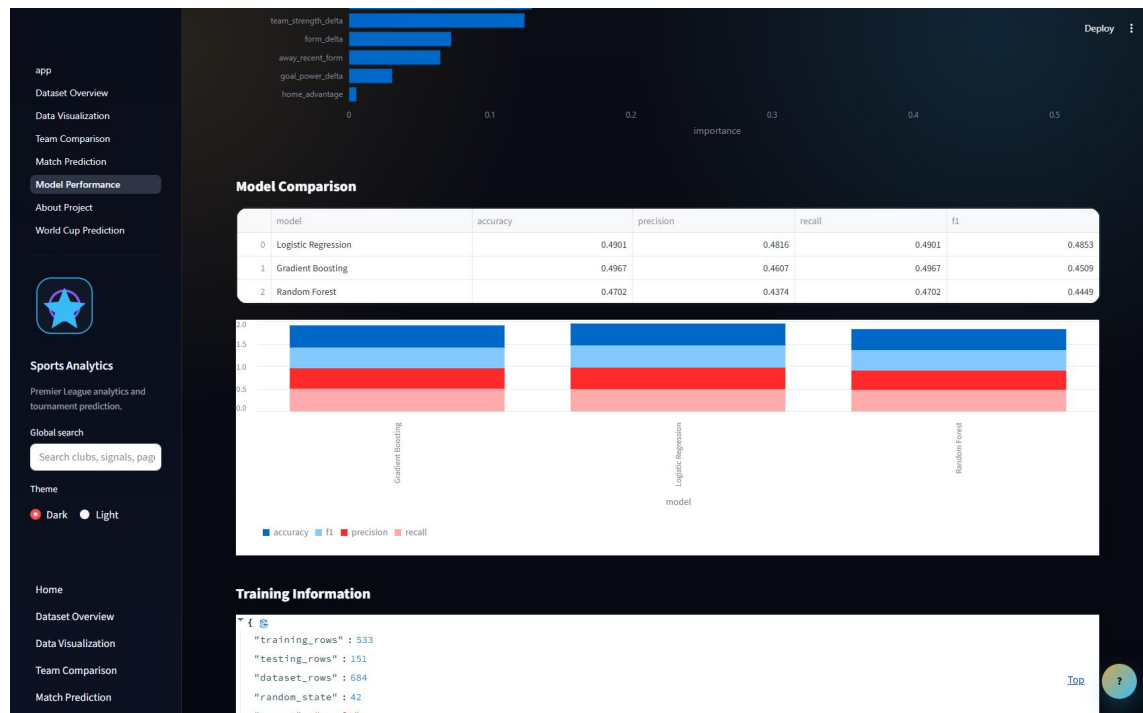


Figure 4.11. Feature importance section from the Model Performance page.

Table 4.3. Confusion matrix from the saved model bundle.

Actual	Away Win	Draw	Home Win
Away Win	26	11	11
Draw	15	9	15
Home Win	11	14	39

4.8 About Page and Project Explanation

Figure 4.12 shows the About Project page. This page explains the project description, technology stack, machine learning pipeline, data source, challenges, future improvements, and university context.

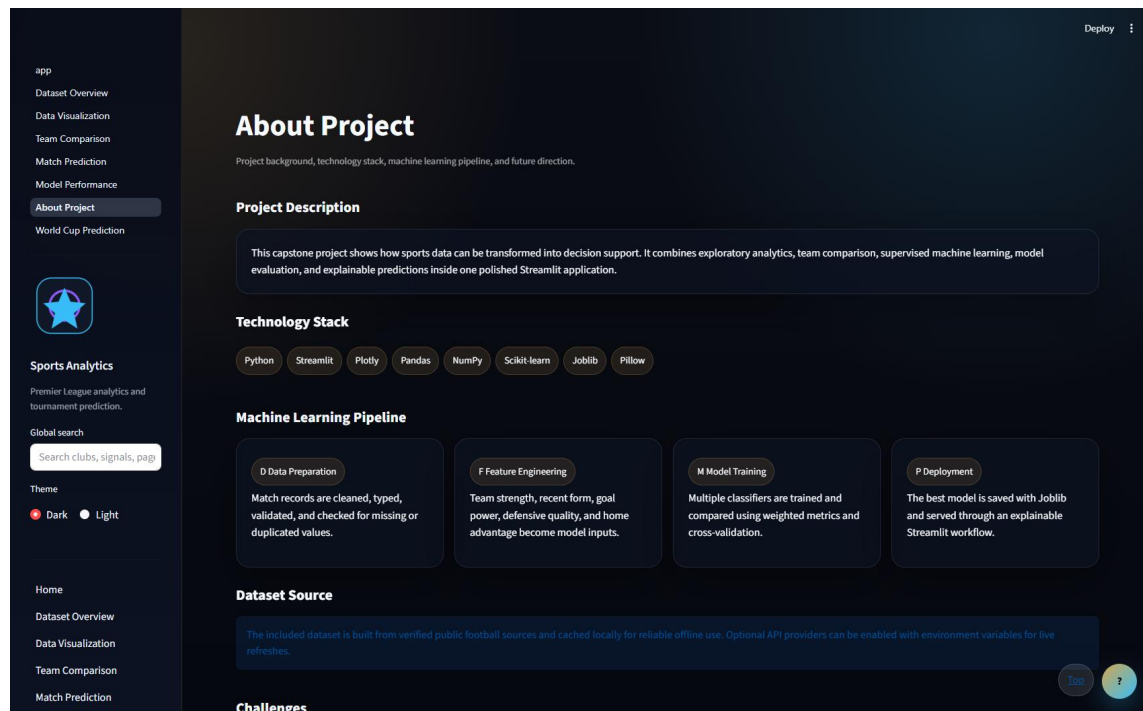


Figure 4.12. About Project page explaining the technology stack and pipeline.

4.9 Discussion

The implementation shows that a complete sports analytics system can be built using Python and Streamlit. The main strength of the system is that it connects real football data, interactive analysis, and prediction output in one application. The dashboard also avoids fake teams and fabricated fixtures by using verified sources, caches, and real fallback lists.

The weakness is that the current model performance is moderate. Football is difficult to predict because results can be affected by tactical changes, player injuries, red cards, weather, and other factors that are not fully represented in the current dataset. The model should therefore be treated as an educational baseline rather than a professional betting system.

CHAPTER 5 CONCLUSION

5.1 Conclusion

This project successfully developed a Sports Analytics Dashboard and Match Outcome Prediction system using machine learning. The application includes dataset overview, visualization, team comparison, match prediction, World Cup prediction, model performance, and project explanation pages. It uses real football data from the OpenFootball public cache and stores a trained model bundle for prediction and evaluation.

The saved model achieved 49.01% accuracy and 48.53% F1 score on the test split. The result is suitable for an undergraduate machine learning dashboard project because it shows the complete process from data collection to deployment, while still reporting the limitations honestly.

5.2 Limitations

- The dataset is limited to the currently cached public match records.
- The model does not include injuries, lineups, tactical formations, weather, or betting market signals.
- Upcoming fixtures depend on verified providers and are not fabricated when unavailable.
- The World Cup prediction uses an indexed strength formula rather than a full tournament simulation.

5.3 Future Work

- Connect live API standings, fixtures, and form when API keys are available.
- Add player-level statistics, injuries, transfers, and expected goals.
- Use stronger time-series validation and more seasons of real match data.
- Deploy the application online with authentication and persistent prediction history.
- Build a mobile-friendly version for easier use during presentations or match analysis.

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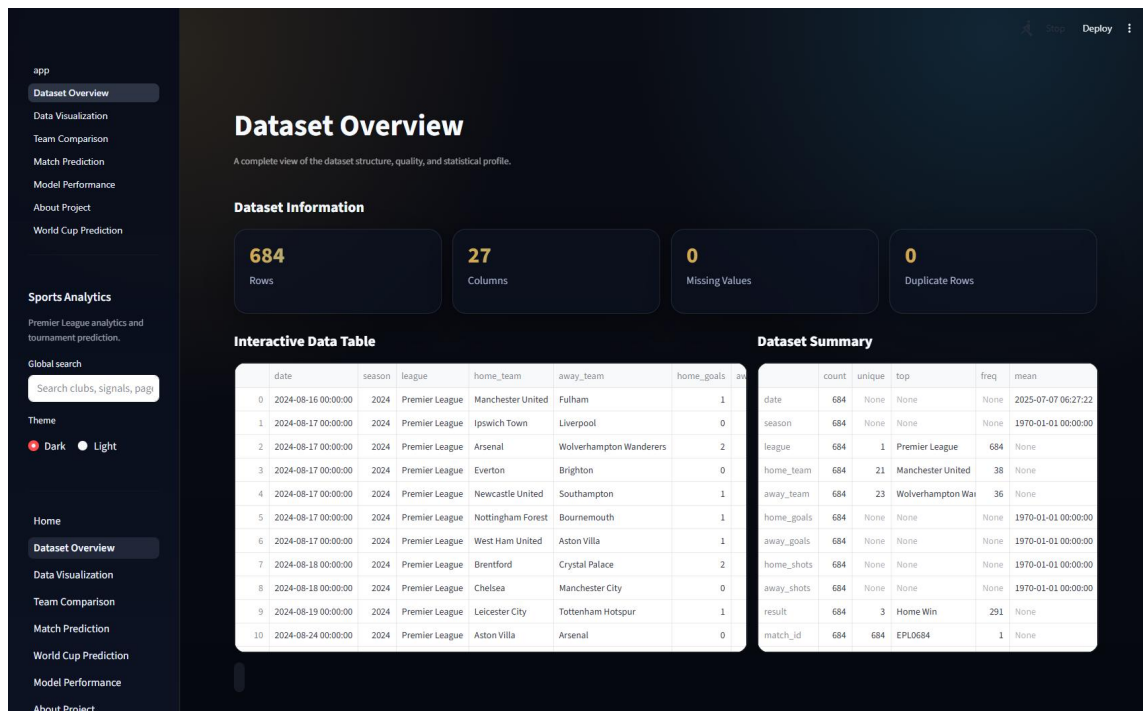
APPENDICES

Appendix A Website Screenshots

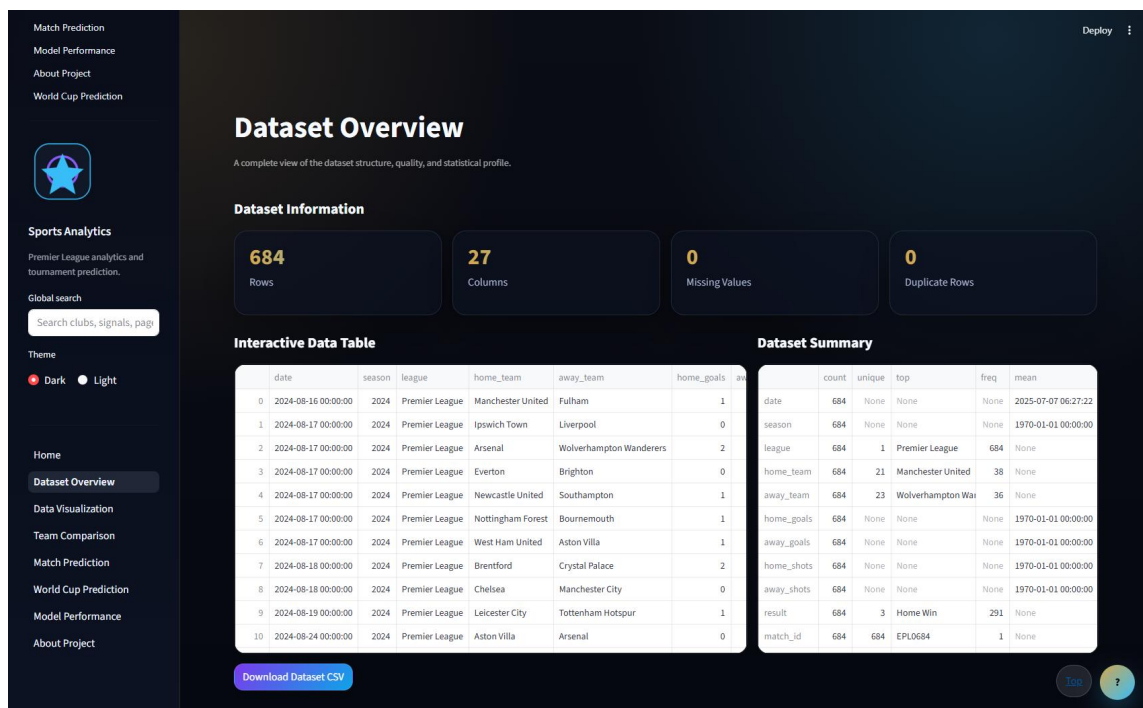
Appendix A contains selected screenshots captured from the actual running Streamlit application. The complete figure set is stored in documentation/report_figures/.



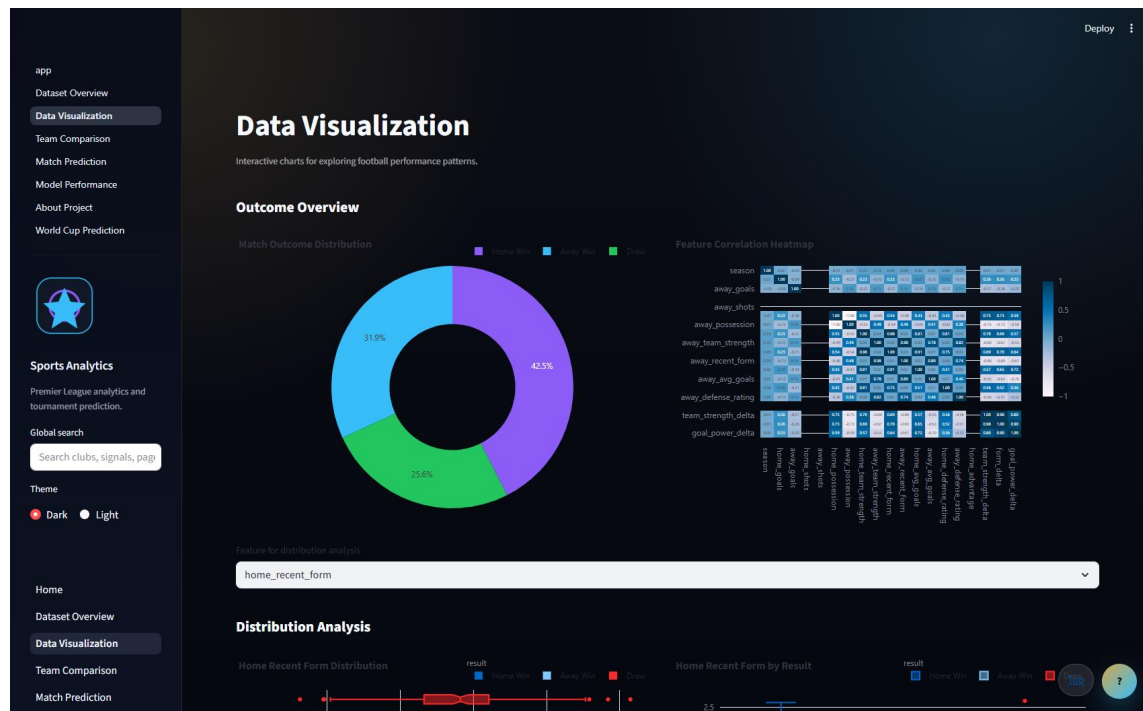
Appendix Figure 4.1. Home page of the Sports Analytics Dashboard.



Appendix Figure 4.2. Dataset Overview page showing dataset statistics and table preview.



Appendix Figure 4.3. Correlation heatmap generated from the football dataset.



Appendix Figure 4.4. Data Visualization page with interactive charts.

Appendix B Important Code Snippets

Table B.1. Important code snippets and their roles.

Area	File	Role
Data loading	utils/data.py	load_dataset() reads dataset/football_matches.csv and parses the date column.
Prediction provider	utils/live_football_data.py	get_real_teams(), get_upcoming_matches(), get_team_statistics(), and predict_match() provide the real-football prediction flow.
Model training	utils/train_model.py	The training script loads verified public data, engineers features, trains models, and saves the Joblib bundle.
World Cup index	utils/world_cup.py	The World Cup page uses weighted signals for official wins, draws, friendly wins, and youth signals.

Appendix C Dataset Description

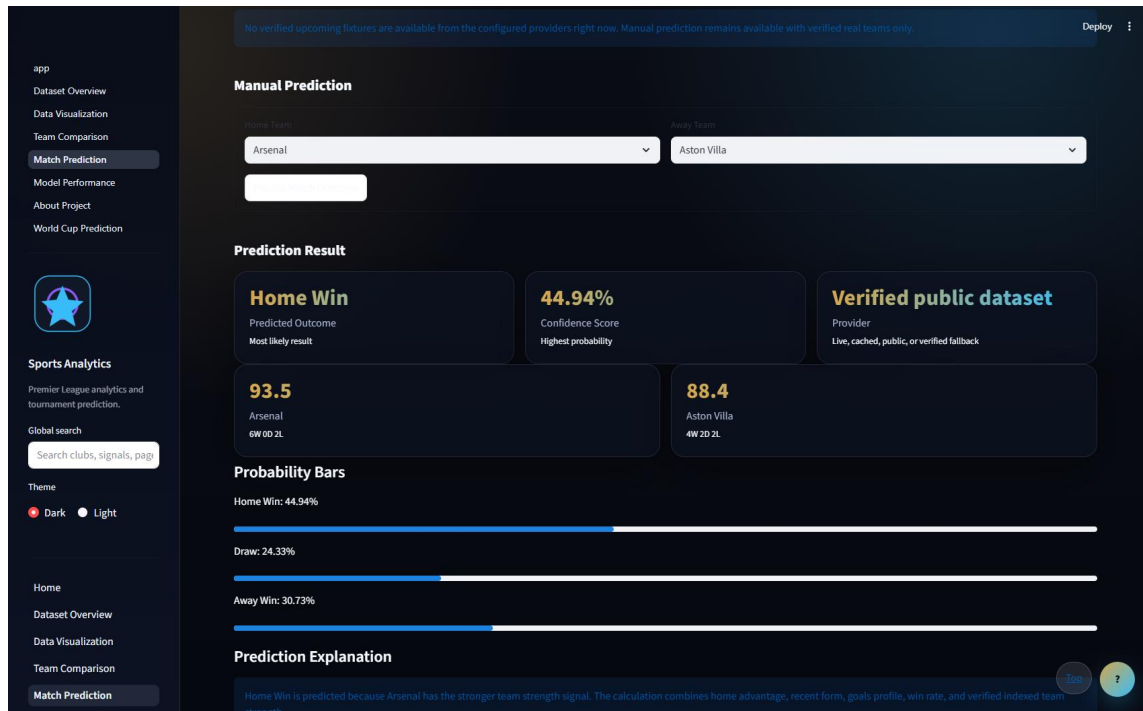
Table C.1. Sample rows from the football dataset.

Date	Home Team	Away Team	Result
2024-08-16	Manchester United	Fulham	Home Win
2024-08-17	Ipswich Town	Liverpool	Away Win
2024-08-17	Arsenal	Wolverhampton Wanderers	Home Win
2024-08-17	Everton	Brighton	Away Win
2024-08-17	Newcastle United	Southampton	Home Win
2024-08-17	Nottingham Forest	Bournemouth	Draw
2024-08-17	West Ham United	Aston Villa	Away Win
2024-08-18	Brentford	Crystal Palace	Home Win

The dataset includes match date, season, league, home team, away team, goals, result, engineered strength features, recent form, and data source.

Appendix D Prediction Examples

Prediction examples are generated inside the Match Prediction page by selecting a league, choosing two verified teams, and pressing Predict Match Outcome. The application stores the prediction history in Streamlit session state and allows the result to be exported as a PDF report.



Appendix Figure D.1. Prediction output example captured from the running application.